TIKTOK TECHJAM 2026 · TRACK 4 (SHOPPING COPILOT)

ShopPilot

An Offline-First Conversational Shopping Copilot for Multi-Turn E-Commerce

Astrid CLI Engine · Zero Token Cost · Offline Default Path

TechnicalScore	Hit Rate@10	MRR	MTTC
0.907	**97.5%**	**0.858**	**2.88 turns**
Starter 0.107 · 8.5× lift	Starter 12.5%	Starter 0.068 · 12.6×	Starter 9.81 · −70.6%

github.com/algorathem/techjam2026-shopping-copilot

01 · PROBLEM

Why Keyword Search Collapses in Conversational Commerce

Single-turn search assumes static intent. Real sessions hit four core failure modes.

01

Aesthetic Under-specification

Vague queries ("something nice for summer") flood pools with flat lexical scores.

02

Semantic Collisions

Polysemy causes category drag ("dress sandals" pulling party dresses).

03

Mid-Session Preference Shifts

Pivots ("actually forget sneakers…") pollute state with stale tokens.

04

Turn Fatigue & Horizon

Abandonment compounds each unnecessary ask; sessions fail at the ≤10-turn limit.

OBJECTIVE FUNCTION

$$\text{TechnicalScore} = 0.50\cdot\text{Hit@10} + 0.30\cdot\text{MRR} + 0.20\cdot((11 - \text{MTTC}) / 10)$$

ShopPilot Engine: In-Memory Multi-Turn Pipeline

Every respo

1. **Intent & Entity Triage**
   Regex triage: buy/browse, audience, taxonomy anchors.

2. **Provenance-Aware DST**
   Slots separate disclosed facts from transient soft prefs.

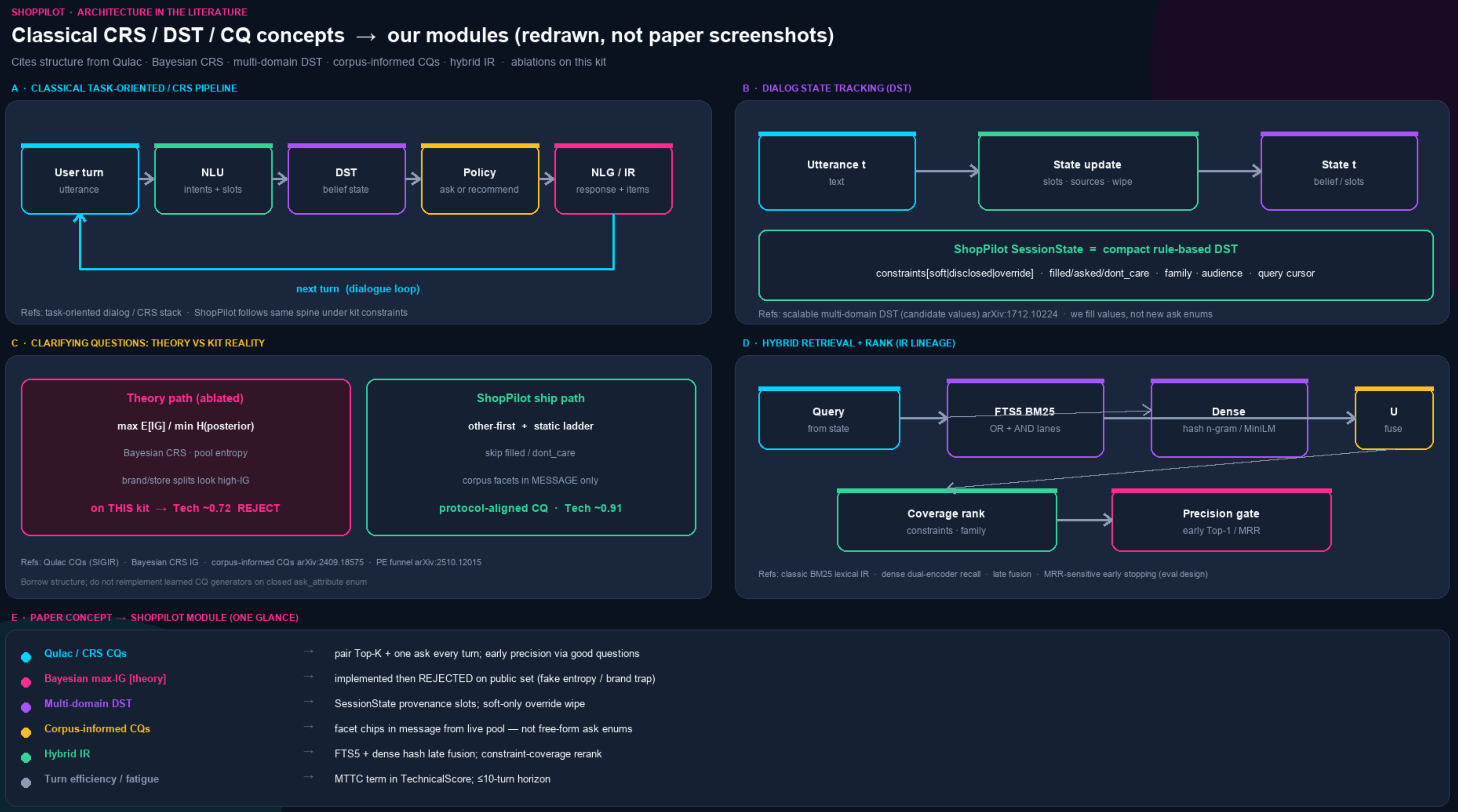
3. **In-Memory Hybrid Recall**
   SQLite FTS5 (BM25) fused with NumPy char n-gram dense hash.

4. **Dynamic Clarification Ladder**
   Asymmetric other-first funnel; skip satisfied slots.

5. **Constraint-Coverage Reranker**
   Exact-match bonuses + leaf-category + audience align.

Classical CRS stack → ShopPilot modules

Redrawn concept diagrams (not paper screenshots) · Qulac · Bayesian CRS · DST · corpus CQs · hybrid IR



State Is the Product: Non-Monotonic Dialog Tracking

Literature spine + TechJam-specific provenance, taxonomy locks, and ask ablations.

CORE INVARIANT · DST belief is cumulative

Turn t3 ("black") ranks against [dress + plus-size + black] — never "black" in isolation.

Selective Soft-Only Invalidation

DST update rule: on override wipe soft prefs; keep disclosed facts the simulator will not re-send.

Taxonomy Boundary Lock

Sense disambiguation at NLU: dress sandals → family:footwear (not garment dresses).

Ask policy vs max-IG

Bayesian max-IG implemented then rejected (Tech ≈0.72). Ship: other-first + static ladder.

Corpus grounding without a CQ model

Facet chips in message from live pool (corpus-informed CQ spirit); enum stays closed.

Official Benchmark: 8.5× Lift Over Baseline

Deterministic evaluator.local_evaluator · Amazon CSJ 50k · N=200 · SHOPPILOT_DENSE=hash

Metric	Weak BM25	ShopPilot (hash)	Delta / Lift
TechnicalScore	0.107	**0.907**	**+0.800 (8.5×)**
Hit Rate@10	12.5%	**97.5%**	**+85.0 pp**
MRR	0.068	**0.858**	**+0.790 (12.6×)**
MTTC (turns)	9.81	**2.88**	**−6.93 (−70.6%)**
LLM Token Cost	0	**0**	**100% offline**

METRIC TRADE-OFF · THE EARLY-TERMINATION PARADOX

Emitting Top-10 too early on Turn 1 risks a low-rank hit (e.g. Rank 8 □ MRR=0.125), freezing a poor score. Because MRR carries 1.5× the weight of MTTC in TechnicalScore, ShopPilot uses Precision Gating (Top-1 limit orders on ambiguous early turns) to gather constraints and force final conversion toward Rank 1.

Performance Breakdown Across Scenarios

High conversion maintained across buying, browse, override, and boundary (N=200).

Scenario	N	Hit@10	MRR	MTTC
Targeted Buying	80	**97.5%**	**0.833**	2.45
Open Browsing	80	**96.3%**	**0.816**	2.79
Intent Override	30	**96.7%**	**0.866**	4.20
Boundary (Don't Care)	10	**100.0%**	**0.933**	3.00
Overall Public Benchmark	**200**	**97.5%**	**0.858**	**2.88**

KEY TAKEAWAY

Preference override incurs a turn penalty (4.20 MTTC) from pivot re-elicitation, but holds 96.7% Hit@10 and 0.866 MRR via clean soft-only state invalidation.

Live Execution: Astrid Multi-Turn Terminal

python3 cli_chat.py --dense hash · /new /state /quit

astrid — dense=hash

```
$ python3 cli_chat.py --dense hash

  Astrid · quiet clarity for every aisle

  You · shoes for my son

  Astrid: Got it (footwear; boys). Any color…?

  □ ask · other

  You · Actually forget sneakers, dress shoes size 9

  Astrid: Updated (footwear; size 9). Color?

  □ soft wipe · disclosed kept · family=footwear

  /state → size=9  asked={other}
```

Vague → Clarify

dress + exploring → ask other → "plus size" locks size (no re-ask).

Taxonomy Lock

"dress sandals" → family=footwear; suppresses garment dresses.

Preference Override

running shoes → dress shoes size 9; soft wipe; size persists.

Bridging Benchmark Metrics to Commercial Value

Kit scores read as merchant KPIs — findability, rank quality, cost-to-serve, infra overhead.

Hit@10 · 97.5%	Minimizes zero-result searches and bounce across sparse catalogs.
MRR · 0.858	Positions converters above the mobile fold without endless scroll.
MTTC · 2.88 turns	Cuts turn fatigue and abandonment before checkout.
Zero Token Footprint	Edge latency, deterministic path, $0 recurring LLM API cost.

08 · CONCLUSION & REPRODUCIBILITY

ShopPilot

Deterministic, Fast, & Open Source

0.907	97.5%	0.858	2.88	0
TechnicalScore	Hit@10	MRR	MTTC	Tokens

ONE-LINE REPRODUCTION

```
export SHOPPILOT_DENSE=hash;  python3 -m evaluator.local_evaluator
```

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Thank you — we welcome your questions.